# Machine Learning Classification of Lung Sound Recordings

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***Abstract*—This report describes a machine learning methodology for classifying lung recordings. The study utilizes the HLS-CMDS dataset, which includes separate heart sound recordings, lung sound recordings, and mixed heart/lung recordings. contains separate heart sound recordings, lung sound recordings, and mixed The workflow consists of audio preprocessing, feature extraction, model training, and parameter tuning. extraction, model training, and parameter tuning, which were done simultaneously using grid search. The best model from grid search was chosen based on the accuracy. Then a comparison of a multi-layer perceptron (MLP) and a convolutional neural network (CNN) was done using macro (average) metrics. MLP beat CNN on all metrics.**

***Index Terms*—machine learning, audio classification, heart sounds, lung sounds, feature extraction, neural networks**

## I. Introduction

The objective of this project is to classify biomedical audio recordings into clinically relevant sound categories using machine learning techniques. Lung sounds are critical diagnostic signals, and automated classification can facilitate rapid screening, efficient data organization, and enhanced diagnostic decision support. This report addresses the challenges associated with accurate classification and summarizes the primary contributions of the study.

- preprocessing and organizing the HLS-CMDS audio dataset;
- extracting useful audio features for classification;
- training and comparing machine learning models;
- evaluating model performance using standard classification metrics.

## II. Dataset

The dataset utilized for this study is derived from the HLS-CMDS dataset, which is organized into HS, LS, and Mix subsets. This analysis focused exclusively on the lung sound recordings from the LS and Mix subsets. In the mixed dataset, 45 lung sounds were shared across all mixed recordings. To eliminate duplicate sounds, a hash string was generated for each recording using the message-digest algorithm (MD5). Each hash string served as a unique fingerprint for its corresponding recording. These fingerprints were used to identify and remove duplicates. Following this process, 94 unique lung sound recordings remained across six classes.

Describe the source of the data, the class labels, the number of samples per class, and any relevant characteristics of the audio files, such as sampling rate, duration, or class imbalance.

TABLE I
Lung Sound Class Distribution

| Label | Recordings |
|---|---|
| Pleural Rub | 18 |
| Rhonchi | 17 |
| Coarse Crackles | 16 |
| Normal | 16 |
| Wheezing | 14 |
| Fine Crackles | 13 |
| **Total** | **94** |

## III. Preprocessing and Feature Extraction

Before model training, the audio recordings were preprocessed to produce a consistent input representation. This section should describe the steps used to clean, normalize, transform, or segment the audio data.

Possible preprocessing steps include:

- loading waveform files and labels;
- resampling audio to a consistent sampling rate;
- trimming or padding recordings to a fixed length;
- normalizing signal amplitude;
- converting audio into features such as MFCCs, spectrograms, or Mel-spectrograms.

If feature extraction was used, include the feature dimensions and explain why those features are appropriate for this classification task.

## IV. Methodology

This section describes the machine learning models and training procedure used for the project. Include enough detail for another person to reproduce the experiment.

### A. Model Architecture

Describe the model or models tested. For example, if a convolutional neural network was used, describe the convolutional layers, activation functions, pooling layers, fully connected layers, dropout, and output layer.

### B. Training Procedure

Describe the train-validation-test split, loss function, optimizer, learning rate, batch size, number of epochs, and any hyperparameter tuning strategy.

### *C. Evaluation Metrics*

The trained models were evaluated using standard classification metrics:

- accuracy;
- precision;
- recall;
- F1-score;
- confusion matrix.

## V. Results

Present the main experimental results in this section. Include a table comparing the tested models or configurations.

TABLE II
Model Performance Comparison

| **Model** | **Accuracy** | **Precision** | **Recall** | **F1-score** |
|---|---|---|---|---|
| Baseline model | TODO | TODO | TODO | TODO |
| Tuned model | TODO | TODO | TODO | TODO |
| Final model | TODO | TODO | TODO | TODO |

Add the most important figures here, such as a confusion matrix, training loss curve, validation accuracy curve, or feature visualization.

Upload `figures/confusion_matrix.png` or replace this placeholder.

Fig. 1. Confusion matrix for the final classification model. Replace the image path with the exported figure from the project.

## VI. Discussion

Discuss what the results mean. Explain which model performed best, which classes were easiest or hardest to classify, and whether the results suggest overfitting, underfitting, or good generalization.

Also describe limitations of the project, such as dataset size, class imbalance, audio quality, limited hyperparameter search, or constraints in preprocessing.

## VII. Conclusion

This project developed a machine learning pipeline for classifying lung sound audio recordings. Summarize the final model, the best performance achieved, and the main lesson learned from the project. End with possible future work, such as collecting more data, improving feature extraction, testing deeper models, or deploying the model in an application.

## Acknowledgment

Optional: thank the instructor, course staff, dataset providers, or teammates. Delete this section if it is not needed.

## References

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